

Images, Functions, and Reflections

Can we generate a random image?



Images and Functions

An image is a function from a canvas to some colors.



(x,y)

$f(x,y)=$



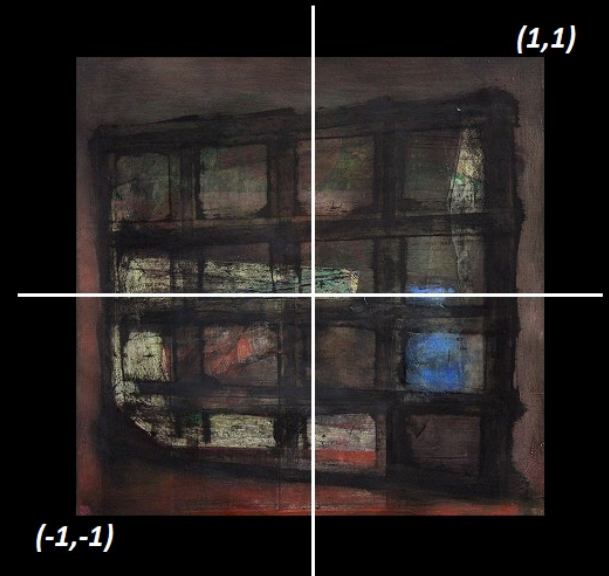
Random image = Random function

Using standard mathematical functions can provide structure which gives the illusion of intent.

How do we build a random function using standard mathematical functions?

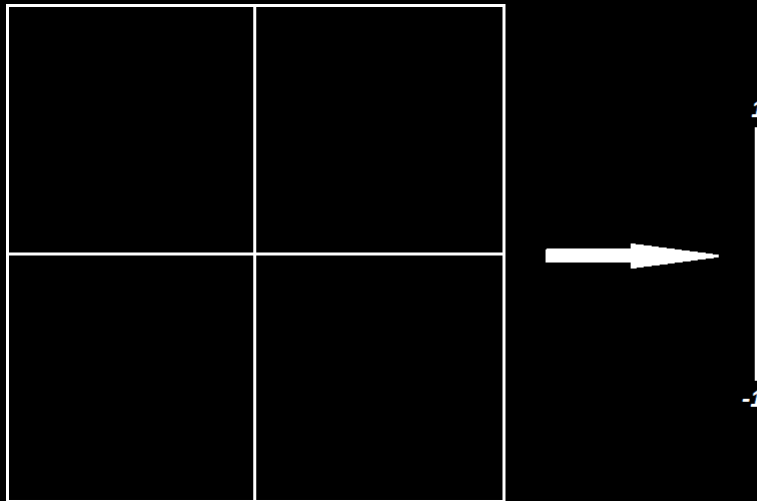
Special Functions

- Impose a coordinate system on the canvas.
 - Place $(1,1)$ at the top right and $(-1,-1)$ at bottom left.
- Call a binary function that maps $[-1,1] \times [-1,1]$ to $[-1,1]$ a *special function*.
- When we compose *special functions*, we get new special functions.



Random Functions

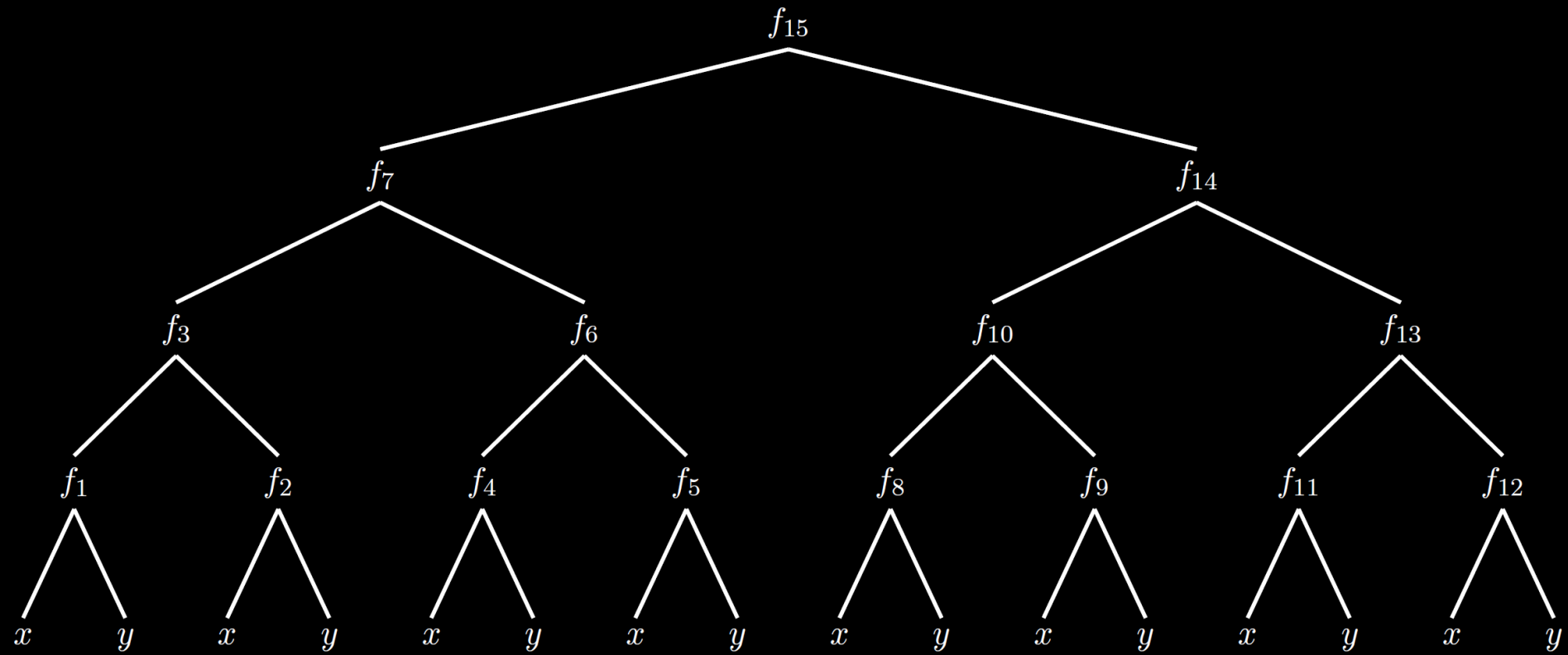
- Make a list of some *special functions*.
- Randomly compose these *special functions*.
- This gives a *randomish* function to $[-1,1]$.



Simple Functions

- $\sin(x \cdot \pi)$
- $\cos(x \cdot \pi)$
- $\sin(2x \cdot \pi)$
- $\cos(2x \cdot \pi)$
- $\sin\left(\pi \cdot (x^2 + y^2)\right)$
- $\text{Avg}(x, y)$
- x
- y
- $x \cdot y$
- x^2
- x^3
- $\min(x, y)$
- $\max(x, y)$

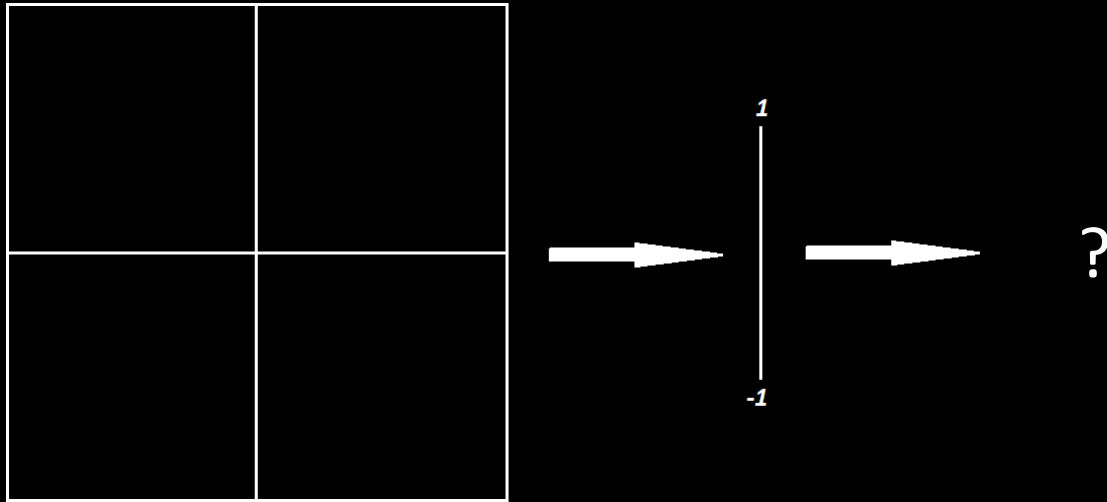
Random Composition



Random Composition

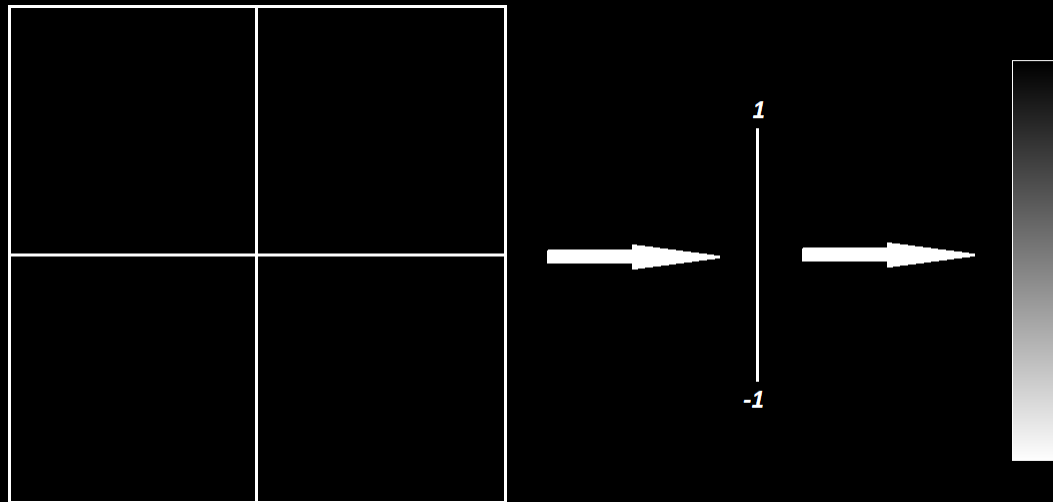
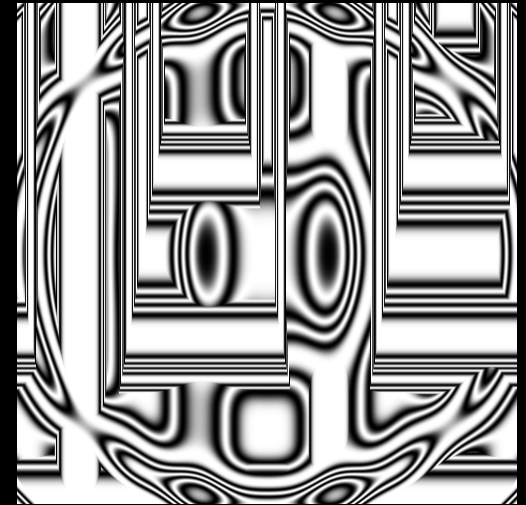
```
procedure randomFunction(height){  
    if (height = 0)  
        return x half the time and y half the time  
    else{  
         $g(x, y) = \text{randomFunction}(\textit{height} - 1)$   
         $h(x, y) = \text{randomFunction}(\textit{height} - 1)$   
         $f(x, y) = \text{specialFunctions.random}()$   
        return( $f(g(x, y), h(x, y))$ )  
    }  
}
```

Random Functions



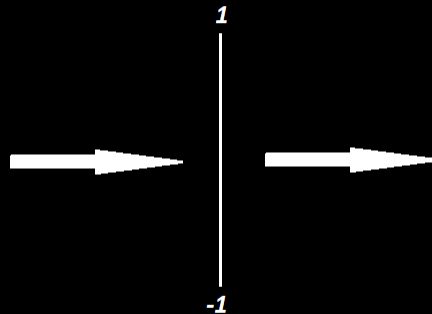
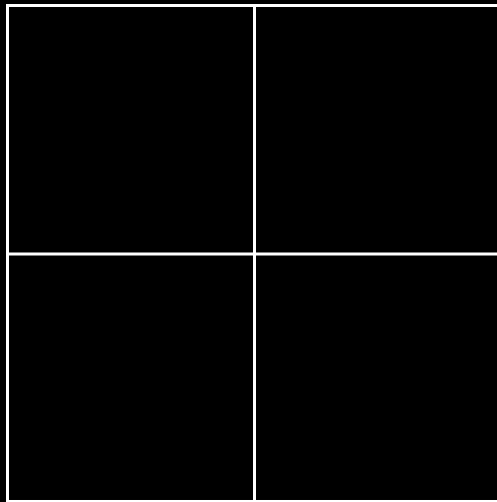
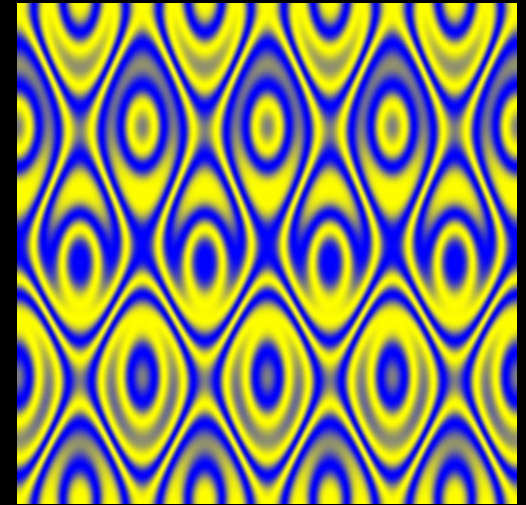
Numbers to Colors

Grayscale



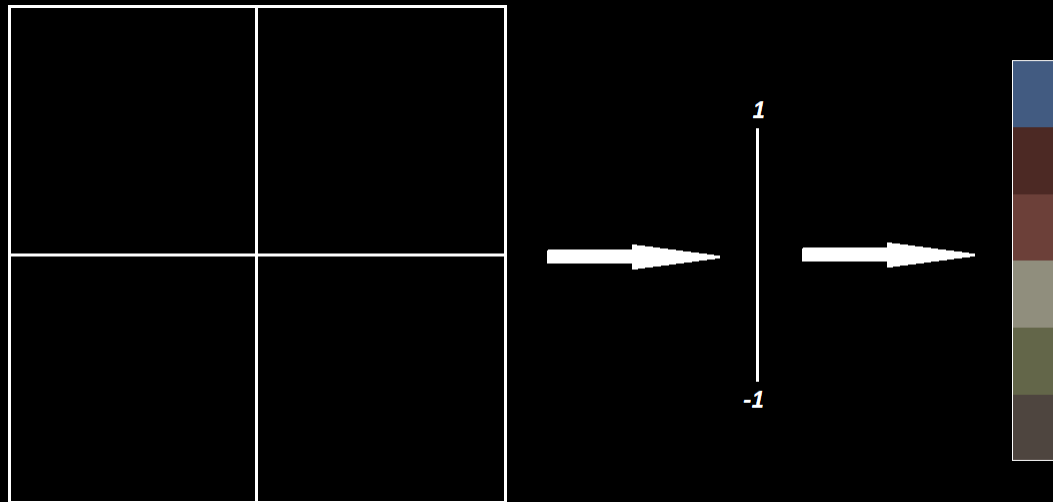
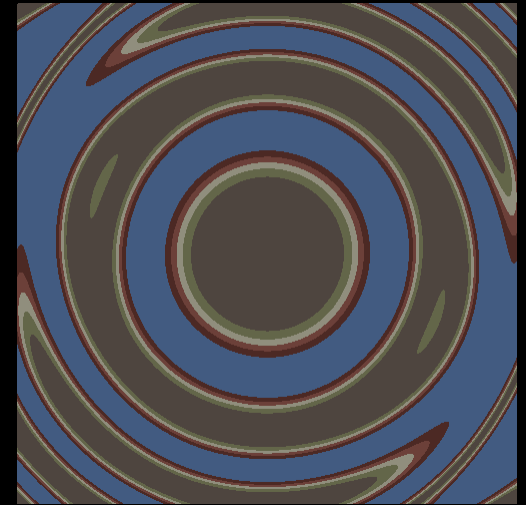
Numbers to Colors

Gradient



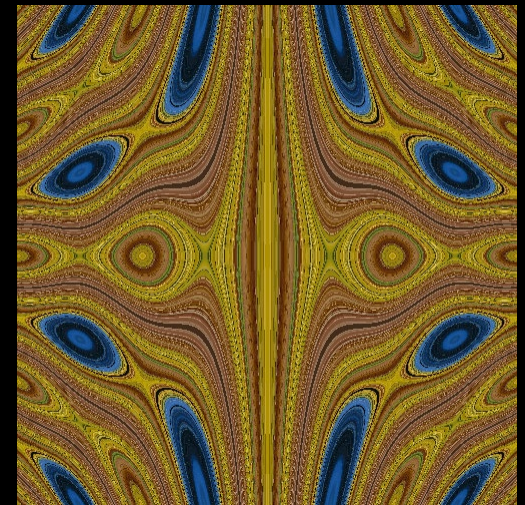
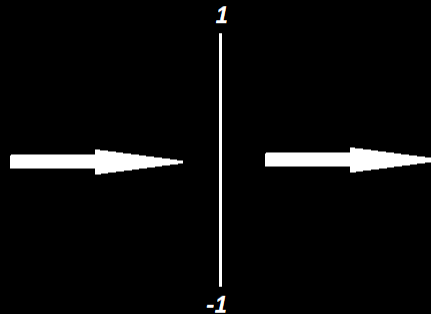
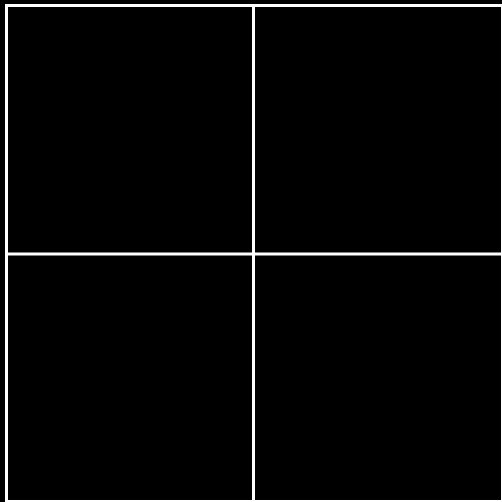
Numbers to Colors

Color array



Numbers to Colors

Interval in another image



Numbers to Colors

Use two functions to map the canvas to another image.

- Generate two special functions $f(x, y)$ and $g(x, y)$.
- Let $F(x, y) = (f(x, y), g(x, y))$.
- The point (x, y) is colored the same as the point $F(x, y)$ in a seed image.

The picture generated is the inverse image of the seed image under the function $F(x, y)$.



Two Functions

Interactive activity

jwsnow.github.io/reflection

More examples